

Footprinting & Scanning Project

Contents

Module1: Footprinting.....	2
1. WHOIS.....	3
SOME KEY FINDINGS.....	3
SOME COMMON ATTACK USES.....	3
STEPS.....	3
CLI OUTPUT.....	4
2. WHATWEB.....	8
SOME KEY FINDINGS.....	9
SOME COMMON ATTACK USES.....	9
STEPS.....	9
CLI OUTPUT.....	9
3. NSLOOKUP.....	10
SOME KEY FINDINGS.....	10
SOME COMMON ATTACK USES.....	10
STEPS.....	11
CLI OUTPUT.....	11
4. CURL -I.....	11
SOME KEY FINDINGS.....	11
SOME COMMON ATTACK USES.....	11
STEPS.....	12
CLI OUTPUT.....	12
5. WAFWOOF.....	13
SOME KEY FINDINGS.....	13
SOME COMMON ATTACK USES.....	13
STEPS.....	13
CLI OUTPUT.....	14
6. DNSRECON.....	15
SOME KEY FINDINGS.....	15
SOME COMMON ATTACK USES.....	15
STEPS.....	15
CLI OUTPUT.....	16
7. Google Hacking Database.....	18

Penetration Testing Report

TASKS.....	18
TROUBLESHOOTING.....	19
8. Maltego (visual OSINT).....	19
SOME KEY FINDINGS.....	20
SOME COMMON ATTACK USES.....	20
TASKS.....	20
9. theHarvester.....	25
SOME KEY FINDINGS.....	25
SOME COMMON ATTACK USES.....	26
TASKS.....	26
CLI OUTPUT.....	27
CLI OUTPUT.....	29
Module 2: Scanning.....	30
1. Zenmap.....	31
SOME KEY FINDINGS.....	31
SOME COMMON ATTACK USES.....	31
TASKS.....	31

Footprinting & Scanning

Footprinting and scanning are core information-gathering phases in ethical hacking used to map a target network and identify potential security flaws before testing begins.

In this project, I carried out footprinting and scanning tasks using an array of appropriate ethical hacking tools.

Module1: Footprinting

Footprinting (or reconnaissance) is the first stage of every real attack. Before touching a target, an attacker quietly builds a complete profile of it using only public information. It refers to the technique used by cybersecurity professionals to identify the exact operating system, software, or hardware configuration of a target network or device. Also known as fingerprinting or banner grabbing, it helps ethical hackers map out a system's structure to uncover known vulnerabilities.

Tools

- whois
- whatweb
- nslookup
- curl

Penetration Testing Report

- wafw00f
- dnsrecon
- Google Hacking Database
- Maltego
- theHarvester

1. WHOIS

I ran **whois** to find the domain registration details. I queried the public domain registration record to find who owns the domain, when it was registered, and its name servers.

SOME KEY FINDINGS

- Registrar (e.g., GoDaddy),
- Name servers (e.g., HostGator),
- Creation/Expiry dates.

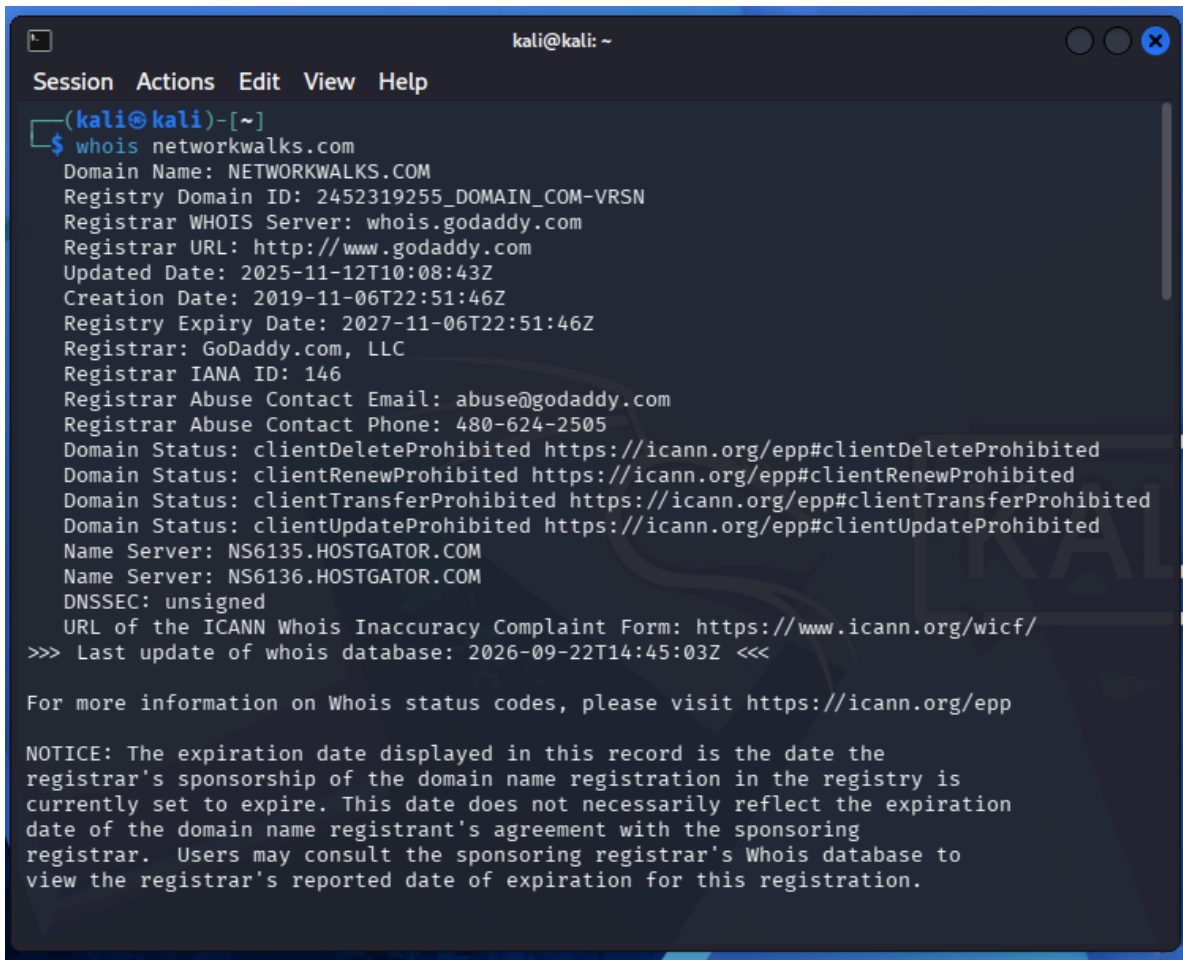
SOME COMMON ATTACK USES

- **Reveals hosting provider:** An attacker would look for known vulnerabilities in that provider.

STEPS

1. Launch the Kali Linux VM; & Open the Terminal;
2. Run the following command: `whois networkwalks.com`

Penetration Testing Report



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ whois networkwalks.com  
Domain Name: NETWORKWALKS.COM  
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.godaddy.com  
Registrar URL: http://www.godaddy.com  
Updated Date: 2025-11-12T10:08:43Z  
Creation Date: 2019-11-06T22:51:46Z  
Registry Expiry Date: 2027-11-06T22:51:46Z  
Registrar: GoDaddy.com, LLC  
Registrar IANA ID: 146  
Registrar Abuse Contact Email: abuse@godaddy.com  
Registrar Abuse Contact Phone: 480-624-2505  
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited  
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited  
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited  
Name Server: NS6135.HOSTGATOR.COM  
Name Server: NS6136.HOSTGATOR.COM  
DNSSEC: unsigned  
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/  
>>> Last update of whois database: 2026-09-22T14:45:03Z <<<  
  
For more information on Whois status codes, please visit https://icann.org/epp  
  
NOTICE: The expiration date displayed in this record is the date the  
registrar's sponsorship of the domain name registration in the registry is  
currently set to expire. This date does not necessarily reflect the expiration  
date of the domain name registrant's agreement with the sponsoring  
registrar. Users may consult the sponsoring registrar's Whois database to  
view the registrar's reported date of expiration for this registration.
```

Figure 1: Screenshot showing the **whois** command return in the Terminal CLI.

CLI OUTPUT

```
(kali@kali)-[~]  
$ whois networkwalks.com  
Domain Name: NETWORKWALKS.COM  
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.godaddy.com  
Registrar URL: http://www.godaddy.com  
Updated Date: 2025-11-12T10:08:43Z  
Creation Date: 2019-11-06T22:51:46Z  
Registry Expiry Date: 2027-11-06T22:51:46Z  
Registrar: GoDaddy.com, LLC  
Registrar IANA ID: 146  
Registrar Abuse Contact Email: abuse@godaddy.com  
Registrar Abuse Contact Phone: 480-624-2505  
Domain Status: clientDeleteProhibited  
https://icann.org/epp#clientDeleteProhibited
```

Penetration Testing Report

CLI OUTPUT

```
Domain Status: clientRenewProhibited
https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited
https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited
https://icann.org/epp#clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form:
https://www.icann.org/wicf/
>>> Last update of whois database: 2026-09-22T14:45:03Z <<<
```

For more information on Whois status codes, please visit
<https://icann.org/epp>

NOTICE: The expiration date displayed in this record is the date the registrar's sponsorship of the domain name registration in the registry is currently set to expire. This date does not necessarily reflect the expiration date of the domain name registrant's agreement with the sponsoring registrar. Users may consult the sponsoring registrar's Whois database to view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois database through the use of electronic processes that are high-volume and automated except as reasonably necessary to register domain names or modify existing registrations; the Data in VeriSign Global Registry Services' ("VeriSign") Whois database is provided by VeriSign for information purposes only, and to assist persons in obtaining information about or related to a domain name registration record. VeriSign does not guarantee its accuracy. By submitting a Whois query, you agree to abide by the following terms of use: You agree that you may use this Data only for lawful purposes and that under no circumstances will you use this Data

Penetration Testing Report

CLI OUTPUT

to: (1) allow, enable, or otherwise support the transmission of mass unsolicited, commercial advertising or solicitations via e-mail, telephone, or facsimile; or (2) enable high volume, automated, electronic processes that apply to VeriSign (or its computer systems). The compilation, repackaging, dissemination or other use of this Data is expressly prohibited without the prior written consent of VeriSign. You agree not to use electronic processes that are automated and high-volume to access or query the Whois database except as reasonably necessary to register domain names or modify existing registrations. VeriSign reserves the right to restrict your access to the Whois database in its sole discretion to ensure operational stability. VeriSign may restrict or terminate your access to the Whois database for failure to abide by these terms of use. VeriSign reserves the right to modify these terms at any time.

The Registry database contains ONLY .COM, .NET, .EDU domains and Registrars.

Domain Name: networkwalks.com

Registry Domain ID: 2452319255_DOMAIN_COM-VRSN

Registrar WHOIS Server: whois.godaddy.com

Registrar URL: https://www.godaddy.com

Updated Date: 2025-11-12T05:08:41Z

Creation Date: 2019-11-06T17:51:46Z

Registrar Registration Expiration Date: 2027-11-06T17:51:46Z

Registrar: GoDaddy.com, LLC

Registrar IANA ID: 146

Registrar Abuse Contact Email: abuse@godaddy.com

Registrar Abuse Contact Phone: +1.4806242505

Domain Status: clientTransferProhibited

<https://icann.org/epp#clientTransferProhibited>

Domain Status: clientUpdateProhibited

<https://icann.org/epp#clientUpdateProhibited>

Domain Status: clientRenewProhibited

<https://icann.org/epp#clientRenewProhibited>

Domain Status: clientDeleteProhibited

Penetration Testing Report**CLI OUTPUT**

```
https://icann.org/epp#clientDeleteProhibited
Registry Registrant ID: Not Available From Registry
Registrant Name: Registration Private
Registrant Organization: Domains By Proxy, LLC
Registrant Street: DomainsByProxy.com
Registrant Street: 100 S. Mill Ave, Suite 1600
Registrant City: Tempe
Registrant State/Province: Arizona
Registrant Postal Code: 85281
Registrant Country: US
Registrant Phone: +1.4806242599
Registrant Phone Ext:
Registrant Fax:
Registrant Fax Ext:
Registrant Email:
https://www.godaddy.com/whois/results.aspx?domain=networkwalks.com&acti
on=contactDomainOwner
Registry Tech ID: Not Available From Registry
Tech Name: Registration Private
Tech Organization: Domains By Proxy, LLC
Tech Street: DomainsByProxy.com
Tech Street: 100 S. Mill Ave, Suite 1600
Tech City: Tempe
Tech State/Province: Arizona
Tech Postal Code: 85281
Tech Country: US
Tech Phone: +1.4806242599
Tech Phone Ext:
Tech Fax:
Tech Fax Ext:
Tech Email:
https://www.godaddy.com/whois/results.aspx?domain=networkwalks.com&acti
on=contactDomainOwner
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
Name Server: NS29.DOMAINCONTROL.COM
Name Server: NS30.DOMAINCONTROL.COM
DNSSEC: unsigned
URL of the ICANN WHOIS Data Problem Reporting System:
http://wdprs.internic.net/
>>> Last update of WHOIS database: 2026-09-22T14:45:15Z <<<
```

Penetration Testing Report

CLI OUTPUT

For more information on Whois status codes, please visit
<https://icann.org/epp>

TERMS OF USE: The data contained in this registrar's Whois database, while believed by the registrar to be reliable, is provided "as is" with no guarantee or warranties regarding its accuracy. This information is provided for the sole purpose of assisting you in obtaining information about domain name registration records. Any use of this data for any other purpose is expressly forbidden without the prior written permission of this registrar. By submitting an inquiry, you agree to these terms and limitations of warranty. In particular, you agree to use this data to allow, enable, or otherwise support the dissemination or collection of data, in part or in its entirety, for any purpose, such as transmission by e-mail, telephone, postal mail, facsimile or other means of mass unsolicited, commercial advertising or solicitation of any kind, including spam. You further agree not to use this data to enable high volume or robotic electronic processes designed to collect or compile this data for any purpose, including mining this data for your own personal or commercial purposes. Failure to comply with the above terms may result in termination of access to the Whois database. These terms may be subject to change at any time without notice.

****NOTICE**** This WHOIS server is being retired. Please use our RDAP service instead.

2. WHATWEB

I ran **whatweb** to fingerprint the web technologies running on the website, e.g., web server, CMS, plugins, frameworks and IP address.

SOME KEY FINDINGS

- HTTP 301,

Penetration Testing Report

- Apache/Nginx version,
- IP address,
- Country,
- Cookies.

SOME COMMON ATTACK USES

- **Exposes exact version numbers:** Attackers collect exact version numbers of running software (e.g., *WordPress 6.1* or *Apache 2.4.41*) and cross-reference them with public vulnerability databases, in order to exploit them.
- **Discovering Sensitive Information:** Attackers can harvest system information like email addresses or user handles embedded within the metadata to launch social engineering or brute-force attacks.

STEPS

1. In the Kali Terminal, run the following command:

```
whatweb networkwalks.com
```

```

kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[U
NITED STATES][US], HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], Redi
rectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by
,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.2], Cookies[__wpdm_client], Count
ry[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HT
TPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenera
tor[WordPress 7.1.2,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Scr
ipt[4684NR-IPIB&pidnVar2=50511&pvtVar2=7&scvVar2=12,application/json,application
/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], Uncomm
onHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx
-cache], WordPress[7.1.2]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.2], Country[UNITED STATES][US],
Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[19
2.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.2,WordPress Download Manager 3
.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&pvtVar2=7
&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascr
ipt], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referr
er-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.2]

```

Figure 2: Screenshot showing the **whatweb** command return in the Terminal CLI.

CLI OUTPUT

```

(kali@kali)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache,

```

Penetration Testing Report

CLI OUTPUT

```
Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache],
HttpOnly[__wpdm_client], IP[192.232.216.135],
RedirectLocation[https://networkwalks.com/],
UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.2],
Cookies[__wpdm_client], Country[UNITED STATES][US],
Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5,
HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135],
jQuery[3.7.1], MetaGenerator[WordPress 7.1.2,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website],
Script[4684NR-IPIB&pidnVar2=50511&pvtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy],
UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.2]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.2],
Country[UNITED STATES][US], Email[info@networkwalks.com], Frame,
Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135],
jQuery[3.7.1], MetaGenerator[WordPress 7.1.2,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website],
Script[4684NR-IPIB&pidnVar2=50511&pvtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy],
UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.2]
```

3. NSLOOKUP

I ran **nslookup** to resolve the **networkwalks.com** domain name to its IP address, using DNS.

SOME KEY FINDINGS

- IP address,
- Mail servers,
- Name servers.

SOME COMMON ATTACK USES

- DNS Reconnaissance and Enumeration: Attackers use **nslookup** to map a target network, locate internal hostnames, identify domain controllers, and find mail exchange (MX) or text (TXT) records to gather intelligence on an organization's infrastructure. They can then implement their findings in phishing attacks.

Penetration Testing Report

STEPS

1. In the Kali Terminal, run the following command:
`nslookup networkwalks.com`

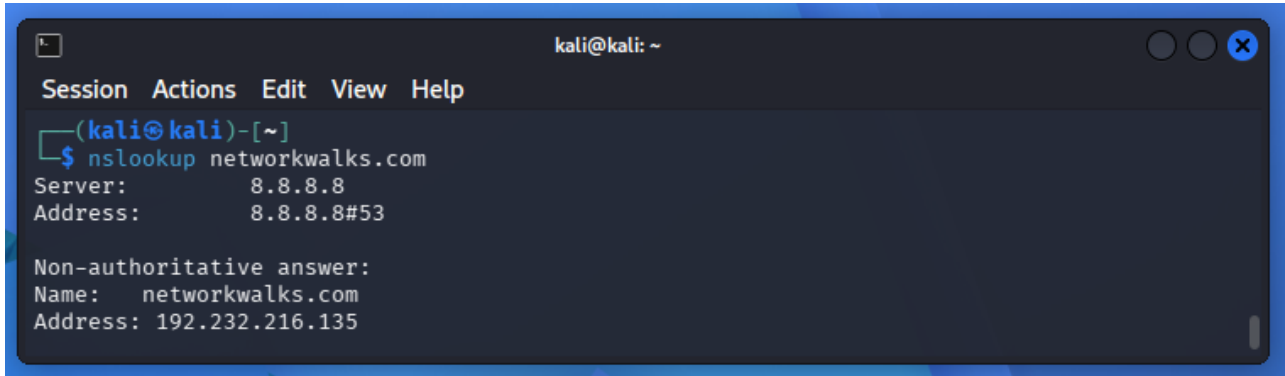
A screenshot of a Kali Linux terminal window. The window title is 'kali@kali: ~'. The menu bar shows 'Session', 'Actions', 'Edit', 'View', and 'Help'. The prompt is '(kali@kali)-[~]'. The command '\$ nslookup networkwalks.com' has been entered. The output shows: 'Server: 8.8.8.8', 'Address: 8.8.8.8#53', 'Non-authoritative answer:', 'Name: networkwalks.com', and 'Address: 192.232.216.135'.

Figure 3: Screenshot showing the **nslookup** command return in the Terminal CLI.

CLI OUTPUT

```
(kali@kali)-[~]  
$ nslookup networkwalks.com  
Server:      8.8.8.8  
Address:     8.8.8.8#53  
  
Non-authoritative answer:  
Name:   networkwalks.com  
Address: 192.232.216.135
```

4. CURL -I

I ran **curl -I** to read the HTTP response headers and then see the server banner, status, cookies and redirects.

SOME KEY FINDINGS

- Server header,
- X-Powered-By,
- security headers.

SOME COMMON ATTACK USES

- **Reconnaissance:** Attackers use flags like **-I** to harvest server information, reveal backend application names and versions, inspect security headers, and map hidden endpoints
- **Missing Headers:** If headers like Content-Security-Policy (CSP), X-Frame-Options, or Strict-Transport-Security (HSTS) are missing, the attacker knows the application is

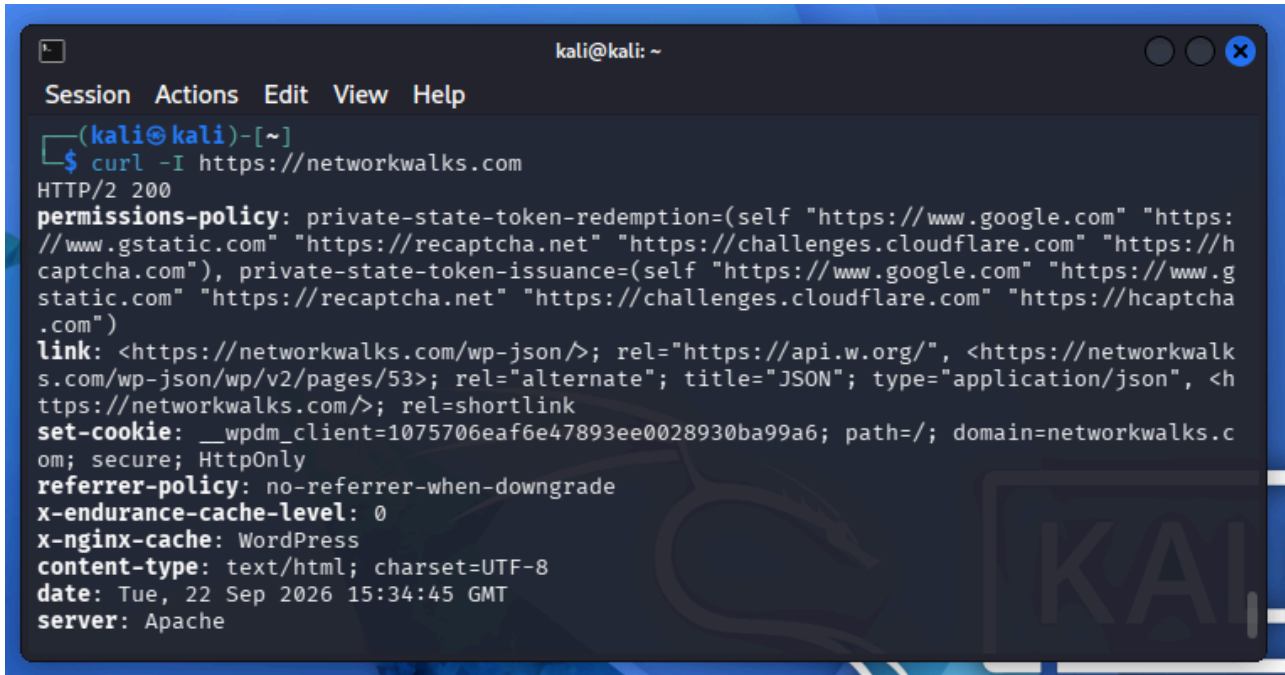
Penetration Testing Report

potentially vulnerable to Cross-Site Scripting (XSS), Clickjacking, or Man-in-the-Middle (MitM) attacks.

STEPS

1. In the Kali Terminal, run the following command:

```
curl -I https://networkwalks.com
```



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
└─$ curl -I https://networkwalks.com  
HTTP/2 200  
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://h  
captcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.g  
static.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcapcha  
.com")  
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalk  
s.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <h  
ttps://networkwalks.com/>; rel=shortlink  
set-cookie: __wpdm_client=1075706eaf6e47893ee0028930ba99a6; path=/; domain=networkwalks.c  
om; secure; HttpOnly  
referrer-policy: no-referrer-when-downgrade  
x-endurance-cache-level: 0  
x-nginx-cache: WordPress  
content-type: text/html; charset=UTF-8  
date: Tue, 22 Sep 2026 15:34:45 GMT  
server: Apache
```

Figure 4: Screenshot showing the **curl -I** command return in the Terminal CLI.

CLI OUTPUT

```
(kali@kali)-[~]  
└─$ curl -I https://networkwalks.com  
HTTP/2 200  
permissions-policy: private-state-token-redemption=(self  
"https://www.google.com" "https://www.gstatic.com"  
"https://recaptcha.net" "https://challenges.cloudflare.com"  
"https://hcaptcha.com"), private-state-token-issuance=(self  
"https://www.google.com" "https://www.gstatic.com"  
"https://recaptcha.net" "https://challenges.cloudflare.com"  
"https://hcaptcha.com")  
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/",  
<https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate";  
title="JSON"; type="application/json", <https://networkwalks.com/>;  
rel=shortlink  
set-cookie: __wpdm_client=1075706eaf6e47893ee0028930ba99a6; path=;
```

Penetration Testing Report

CLI OUTPUT

```
domain=networkwalks.com; secure; HttpOnly  
referrer-policy: no-referrer-when-downgrade  
x-endurance-cache-level: 0  
x-nginx-cache: WordPress  
content-type: text/html; charset=UTF-8  
date: Tue, 22 Sep 2026 15:34:45 GMT  
server: Apache
```

5. WAFWOOF

I ran **wafw00f** to detect whether a Web Application Firewall (WAF) was protecting the target site.

SOME KEY FINDINGS

- "Behind ModSecurity / Cloudflare / Sucuri" or "No WAF"

SOME COMMON ATTACK USES

- **Firewall Protection Fingerprinting:** This tool is used by attackers and penetration testers to detect and fingerprint Web Application Firewalls (WAFs) protecting a target website.

STEPS

1. In the Kali Terminal, run the following command:
`wafw00f networkwalks.com`

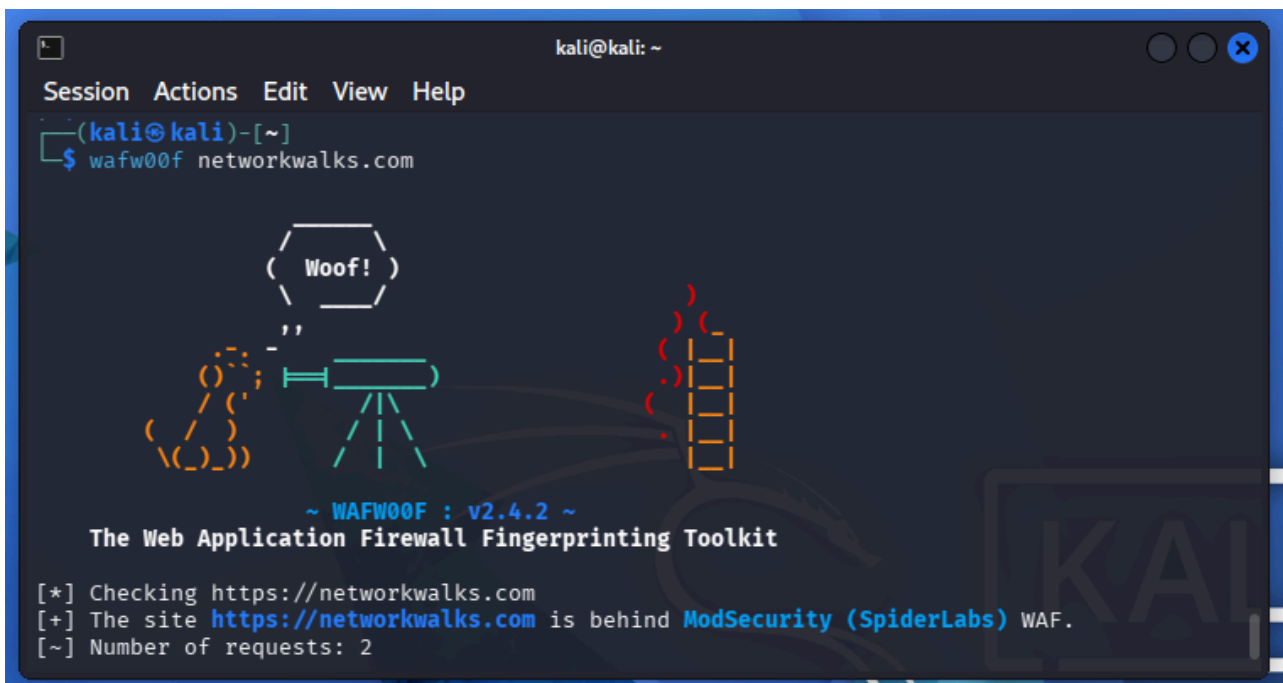


Figure 5: Screenshot showing the **wafw00f** command return in the Terminal CLI.

Penetration Testing Report

CLI OUTPUT

```

└─(kali㉿kali)-[~]
└─$ wafw00f networkwalks.com

      /-----\
     (  Woof!  )
      \  ____/
        '
      .-. -
     ( ) `; |==|-----)
      / ( '      /|\
     ( / )      / | \
    \(-)-)      /  |  \

                                     )
                                   ) ( _
                                   ( |__|
                                   .) |__|
                                   (  |__|
                                   .  |__|
                                   |__|

~ WAFW00F : v2.4.2 ~
The Web Application Firewall Fingerprinting Toolkit

[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity
(SpiderLabs) WAF.
[~] Number of requests: 2

```

6. DNSRECON

I ran **dnsrecon** to enumerate all DNS records, e.g., name servers, mail servers, SPF, TXT and service (SRV) records.

SOME KEY FINDINGS

- A records,
- MX records,
- NS records,
- subdomains.

SOME COMMON ATTACK USES

- **DNS Footprint Mapping:** Attackers use this tool to extract mail servers, DNS software version (Bind 9.16.23), SPF policy and cPanel service records. Each record potentially helps an attacker understand the email and hosting setup.

STEPS

1. In the Kali Terminal, run the following command:

```
dnsrecon -d networkwalks.com
```

Penetration Testing Report

```

kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ dnsrecon -d networkwalks.com
2026-09-22T11:56:24.876981-0400 INFO Starting enumeration for domain: networkwalks.com
2026-09-22T11:56:24.877459-0400 INFO std: Performing General Enumeration against: networkwalks.com ...
2026-09-22T11:56:25.649921-0400 ERROR No answer for DNSSEC query for networkwalks.com
2026-09-22T11:56:29.970080-0400 INFO NS ns6135.hostgator.com 50.87.144.87
2026-09-22T11:56:30.552827-0400 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-09-22T11:56:30.553285-0400 INFO NS ns6136.hostgator.com 192.232.216.131
2026-09-22T11:56:31.099445-0400 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-09-22T11:56:32.199237-0400 INFO MX mail.networkwalks.com 192.232.216.135
2026-09-22T11:56:32.370779-0400 INFO A networkwalks.com 192.232.216.135
2026-09-22T11:56:33.560109-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-09-22T11:56:33.560305-0400 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-09-22T11:56:34.288089-0400 INFO Enumerating SRV Records
2026-09-22T11:56:41.746507-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.14 443
2026-09-22T11:56:41.747878-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.11 443
2026-09-22T11:56:41.748926-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.14 443
2026-09-22T11:56:41.749799-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.15 443
2026-09-22T11:56:41.751026-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.15 443
2026-09-22T11:56:41.751818-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.11 443
2026-09-22T11:56:41.752345-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.9 443
2026-09-22T11:56:41.752786-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.9 443
2026-09-22T11:56:41.755744-0400 INFO 8 Records Found
2026-09-22T11:56:41.756841-0400 INFO Completed enumeration for domain: networkwalks.com

```

Figure 6: Screenshot showing the **dnsrecon** command return in the Terminal CLI.

CLI OUTPUT

```

(kali@kali)-[~]
$ dnsrecon -d networkwalks.com
2026-09-22T11:56:24.876981-0400 INFO Starting enumeration for domain: networkwalks.com
2026-09-22T11:56:24.877459-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-09-22T11:56:25.649921-0400 ERROR No answer for DNSSEC query for networkwalks.com
2026-09-22T11:56:29.970080-0400 INFO NS ns6135.hostgator.com 50.87.144.87
2026-09-22T11:56:30.552827-0400 INFO Bind Version for 50.87.144.87

```

Penetration Testing Report

CLI OUTPUT

```
"9.16.23-RH"
2026-09-22T11:56:30.553285-0400 INFO      NS ns6136.hostgator.com
192.232.216.131
2026-09-22T11:56:31.099445-0400 INFO      Bind Version for
192.232.216.131 "9.16.23-RH"
2026-09-22T11:56:32.199237-0400 INFO      MX mail.networkwalks.com
192.232.216.135
2026-09-22T11:56:32.370779-0400 INFO      A networkwalks.com
192.232.216.135
2026-09-22T11:56:33.560109-0400 INFO      TXT networkwalks.com v=spf1 +a
+mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-09-22T11:56:33.560305-0400 INFO      TXT networkwalks.com
google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-09-22T11:56:34.288089-0400 INFO      Enumerating SRV Records
2026-09-22T11:56:41.746507-0400 INFO      SRV
_autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net
184.94.204.14 443
2026-09-22T11:56:41.747878-0400 INFO      SRV
_autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net
184.94.203.11 443
2026-09-22T11:56:41.748926-0400 INFO      SRV
_autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net
184.94.203.14 443
2026-09-22T11:56:41.749799-0400 INFO      SRV
_autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net
184.94.203.15 443
2026-09-22T11:56:41.751026-0400 INFO      SRV
_autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net
184.94.204.15 443
2026-09-22T11:56:41.751818-0400 INFO      SRV
_autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net
184.94.204.11 443
2026-09-22T11:56:41.752345-0400 INFO      SRV
_autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net
184.94.203.9 443
2026-09-22T11:56:41.752786-0400 INFO      SRV
_autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net
184.94.204.9 443
2026-09-22T11:56:41.755744-0400 INFO      8 Records Found
2026-09-22T11:56:41.756841-0400 INFO      Completed enumeration for domain:
networkwalks.com
```

Penetration Testing Report

7. Google Hacking Database

TASKS

- Find 10 live vulnerable security camera links that are exposed & accessible from the Internet. List them in the table below along with their relevant dork that you have used.

No.	Link	Relevant Dork	Username /Password (if any)
1	https://80.152.138.183/ViewerFrame?Mode=Motion&Language=0	<code>inurl:/ViewerFrame?intitle:"Network Camera NetworkCamera"</code>	No
2	https://webcam.io/users/sign_in	<code>intitle:"webcam" inurl:login</code>	Yes
3	http://95.255.183.164:8080/multi.html	<code>intitle:"WEBCAM 7"-inurl:/admin.html</code>	No
4	http://62.202.21.238:8081/	<code>intitle:"WEBCAM 7"-inurl:/admin.html</code>	Yes
5	http://109.233.191.130:8080/	<code>intitle:"webcamXP 5" inurl:8080 'Live'</code>	No
6	http://85.93.53.175:8080/home.html	<code>intitle:"webcamXP 5" inurl:8080 'Live'</code>	No
7	http://193.77.49.122/	<code>intitle:"IP CAMERA Viewer" intext:"setting Client setting"</code>	No
8	http://99.114.240.169:8080/	<code>intitle:"webcam 7" inurl:'/gallery.html'</code>	No
9	http://176.106.38.115:82/Pages/login.htm?0.17693531443781252	<code>intitle:"NoVus IP camera" -com</code>	Yes
10	https://www.skylinewebcams.com/webcam/italia/lazio/roma/fontana-di-trevi.html	<code>inurl:webcam site:skylinewebcams.com inurl:roma</code>	No

- Find 10 listings which contain downloadable mathematics ebooks in PDF format:

Penetration Testing Report

No.	Link	Relevant Dork	Username /Password (if any)
1	https://www.jsoftware.com/books/pdf/	intitle:index.of "parent directory" mathematics pdf	No
2	https://ochicken.net/library/Mathematics/	intitle:index.of "parent directory" mathematics pdf	No
3	https://www.unm.edu/~megrad/Math/	intitle:index.of "parent directory" mathematics pdf	No
4	http://erewhon.superkuh.com/library/Math/	intitle:index.of "parent directory" mathematics pdf	No
5	https://www.aetkin.com/files/Math%20150%20Calculus%20I/Advanced%20Calculus%20Textbook/	intitle:index.of "parent directory" advanced mathematics pdf	No
6	https://lira.epac.to/DOCS-TECH/Math/Engineering%20and%20Applied/	intitle:index.of "parent directory" advanced mathematics pdf	No
7	https://education.giakonda.org.uk/Maths/	intitle:index of "parent directory" books math pdf	No
8	https://www.math.uchicago.edu/~may/BOOKS/	intitle:index of "parent directory" books math pdf	No
9	https://isidore.co/misc/Physics%20papers%20and%20books/Mathematics/	intitle:index of "parent directory" books math theory pdf	No
10	https://math.uchicago.edu/~margalit/repthy/	intitle:index of "parent directory" books math theory pdf	No

TROUBLESHOOTING

1. The Google Hacking Database URL refused to open on my browser.
 - a. **Solution:** I ran a VPN and tunneled into a US server.

8. Maltego (visual OSINT)

Maltego is a graphical analysis and open-source intelligence (OSINT) tool used to gather, connect, and visualize data for cyber investigations.

SOME KEY FINDINGS

- Subdomains.

Penetration Testing Report

- Email domains.

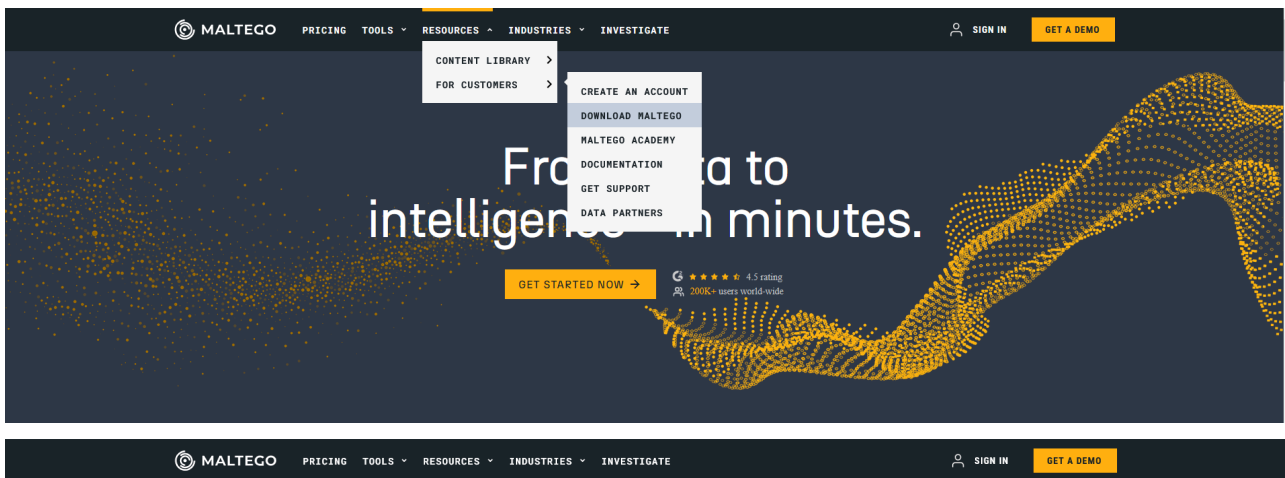
SOME COMMON ATTACK USES

- **Reveals email domains:** An attacker would look for email domains in order to launch tailored social engineering attacks.

TASKS

This tool was installed, configured and used to fingerprint the **networkwalks.com** domain in the following steps:

1. Download, install and setup Maltego on Windows.
 - a. Open the website **<https://maltego.com>** and download Maltego



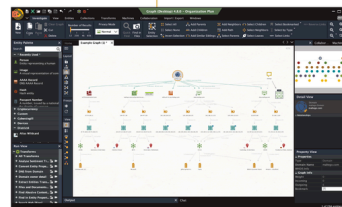
DOWNLOAD

Maltego Graph for Windows

Get everything you need to speed up complex investigations from hours to minutes.

Operating System Windows	Extension .exe + Java (x64)
DOWNLOAD MALTEGO GRAPH	

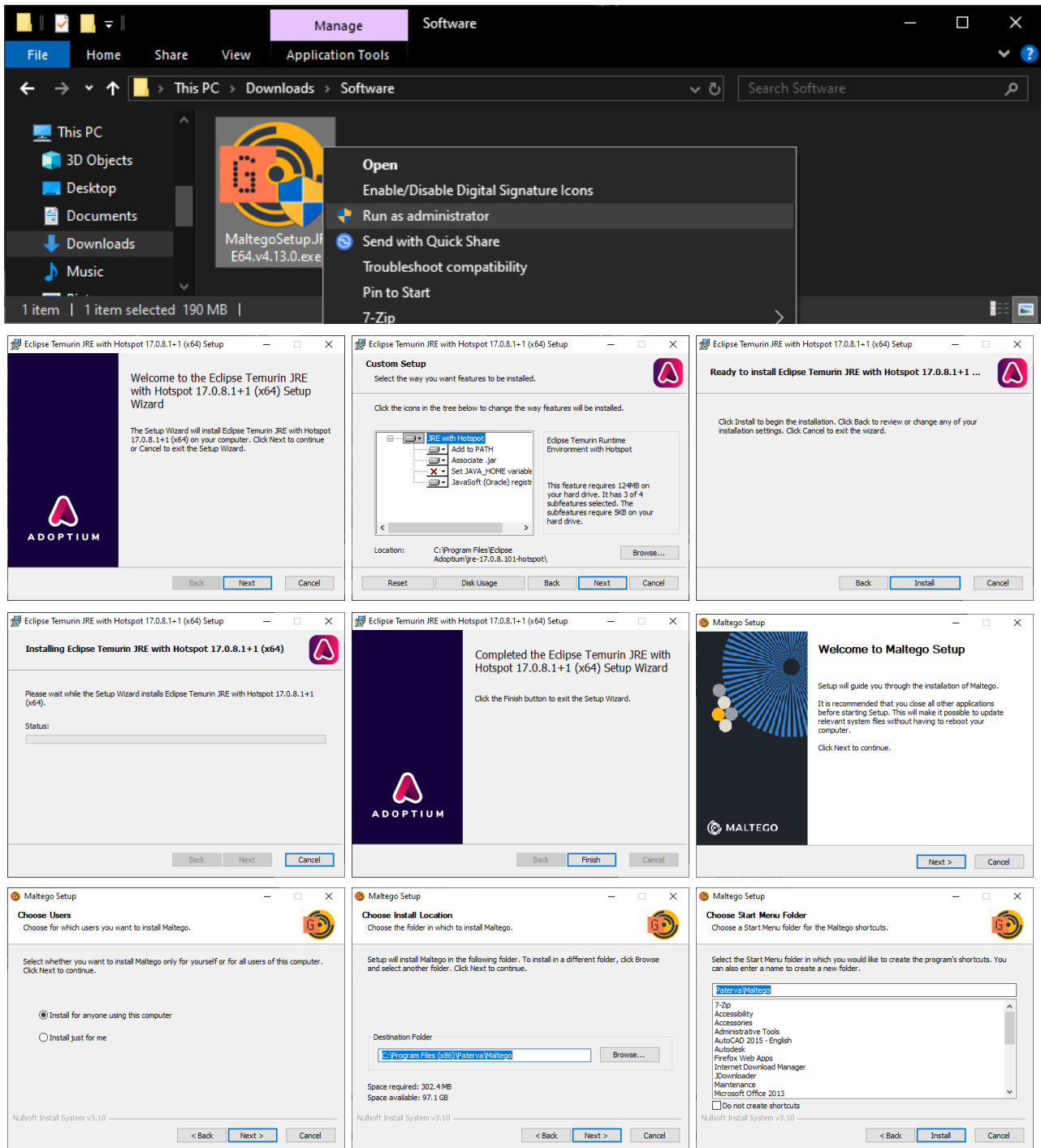
Version JRE64.4.13.0 [What's new](#)



- b. Run the **Maltego** installer.

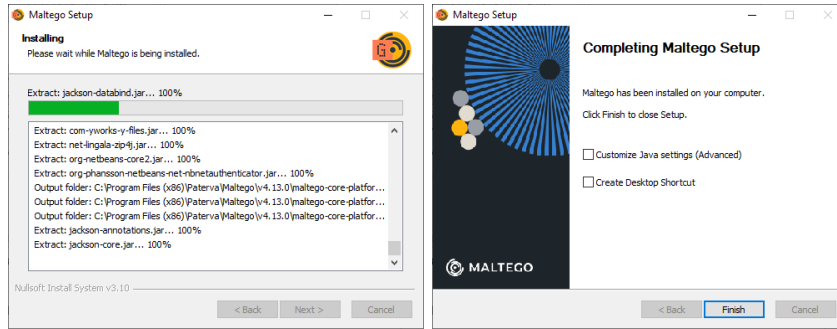
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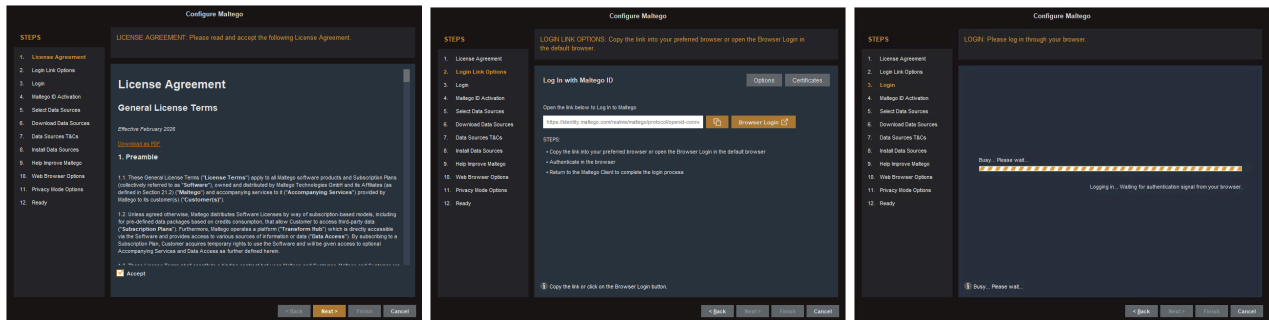
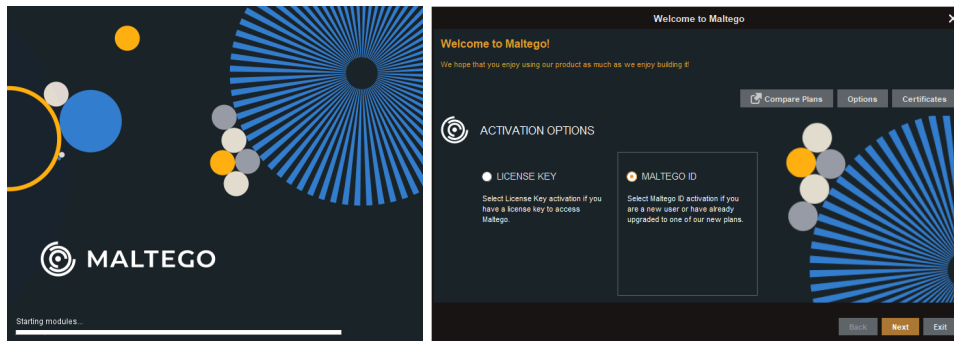


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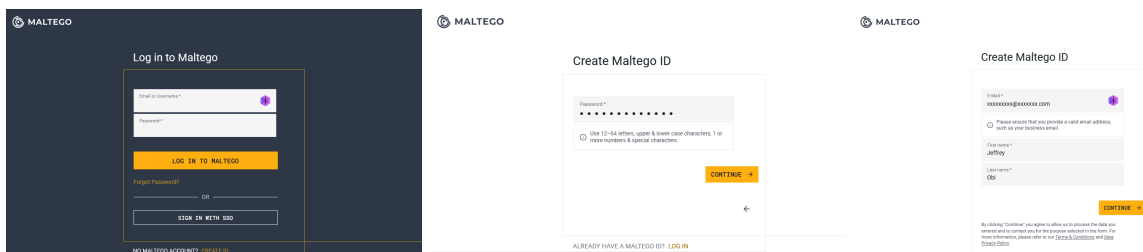
Penetration Testing Report



c. Run Maltego after installation & complete the configuration.



d. Complete the free Maltego account creation and then login.



Penetration Testing Report



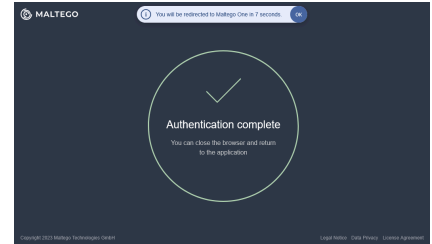
e. Fill the form & complete your registration to get your free Maltego ID.

Complete Your Profile

The screenshot shows a form titled "Penetration Testing" with the following fields:

- Organization Name *
- Organization Size * (dropdown menu with "1-10" selected)
- Your Role *
- Organization website (e.g. www.company.com)
- How did you first learn about Malware? (dropdown menu with "Training" selected)
- Email of Business Unit *

At the bottom of the form is a "CONTINUE" button with a right arrow.



f. Return to Maltego & complete the configuration.

Configure Mailboxes

STEPS

1. License Agreement
2. Login Link Option
3. Login
4. Mailbox E Activation
5. Select Data Sources
6. Download Data Sources
7. Data Sources Table
8. Install Data Sources
9. Help Improve Mailbox
10. Web Browser Update
11. Privacy Policy Center
12. Ready

INSTALL DATA SOURCES

A summary of the progress to install items from the chosen Data Sources is shown below.

Complete

166 Mailbox items were installed.

22 Applications	Service
238 Transmits	
107 Users	
245 Frames	
48 Transceiver Sets	
8 Machines	

◀ Back

Next ▶

Finish

Cancel

STEPS

1. License Agreement

2. Log-in Link Options

3. Log-in

4. Malware G. Activation

5. Social Data Sources

6. Download Data Sources

7. Domain Sources TACs

8. Social Data Sources

9. Map Importing Sources

10. Web Crawler Options

11. Privacy Mode Options

12. Ready

Configure Maltego

PRIVACY MODE OPTIONS: Select the Privacy Mode to allow or limit web requests.

Privacy Mode

an Normal

Normal Privacy Mode: Use this mode for richest Maltego experience. Your Maltego Desktop is allowed to fetch some data directly from the internet. Examples include loading of entity images from URLs and Favicons for available websites.

Stealth Privacy Mode: Use this mode if you are an investigator and it is important for you to remain traffic, originating from your computer(s) to Your Maltego Desktop and indirectly any data sent to the Internet. Downloading of entity images and image overlays will be blocked, from that transforms might still make direct connections to the target(s) from the host(s) inside or from your computer (in case of local transforms).

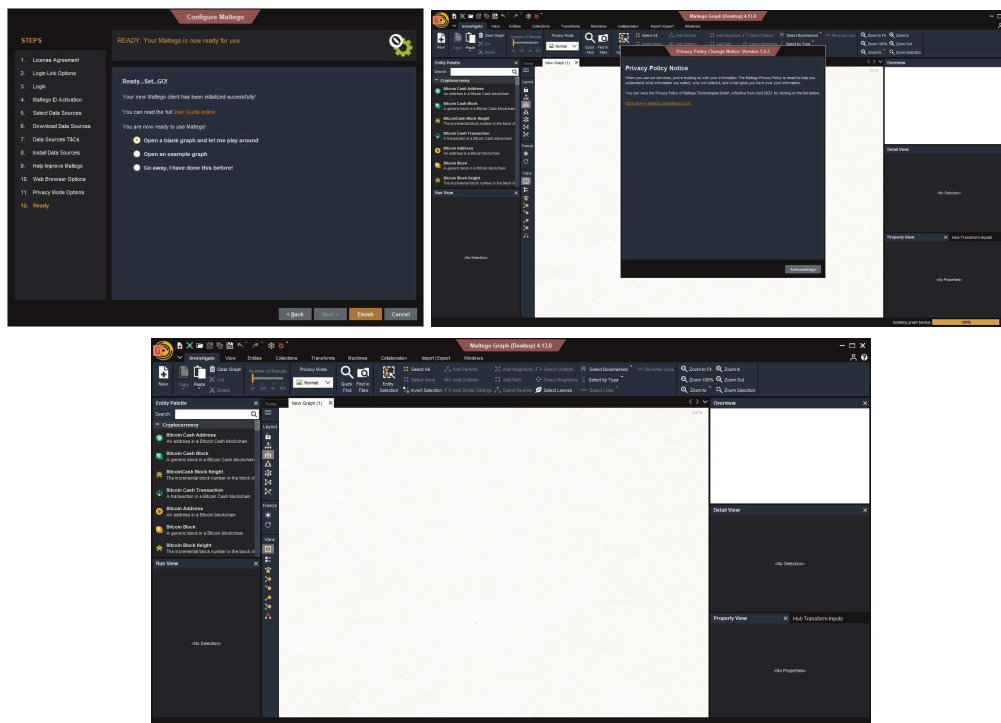
Back

Next >

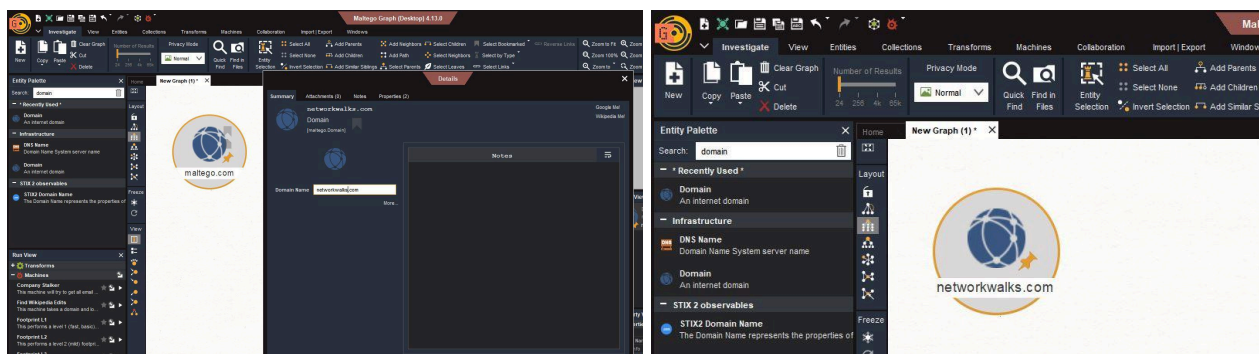
Cancel

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Penetration Testing Report



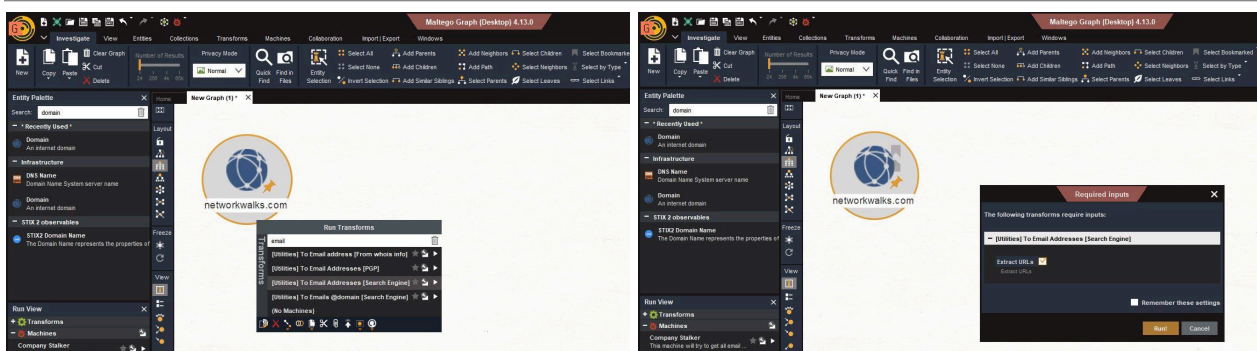
2. I found all email addresses related to the target organization domain networkwalks.com.
 - a. Search “**domain**” in the **Entity Palette** panel and drag the **Domain** tile into the **New Graph** window. In the **New Graph** window, double-click on the new Domain entity & change the ‘**Domain Name**’ entry from “**maltego.com**” to “**networkwalks.com**.”



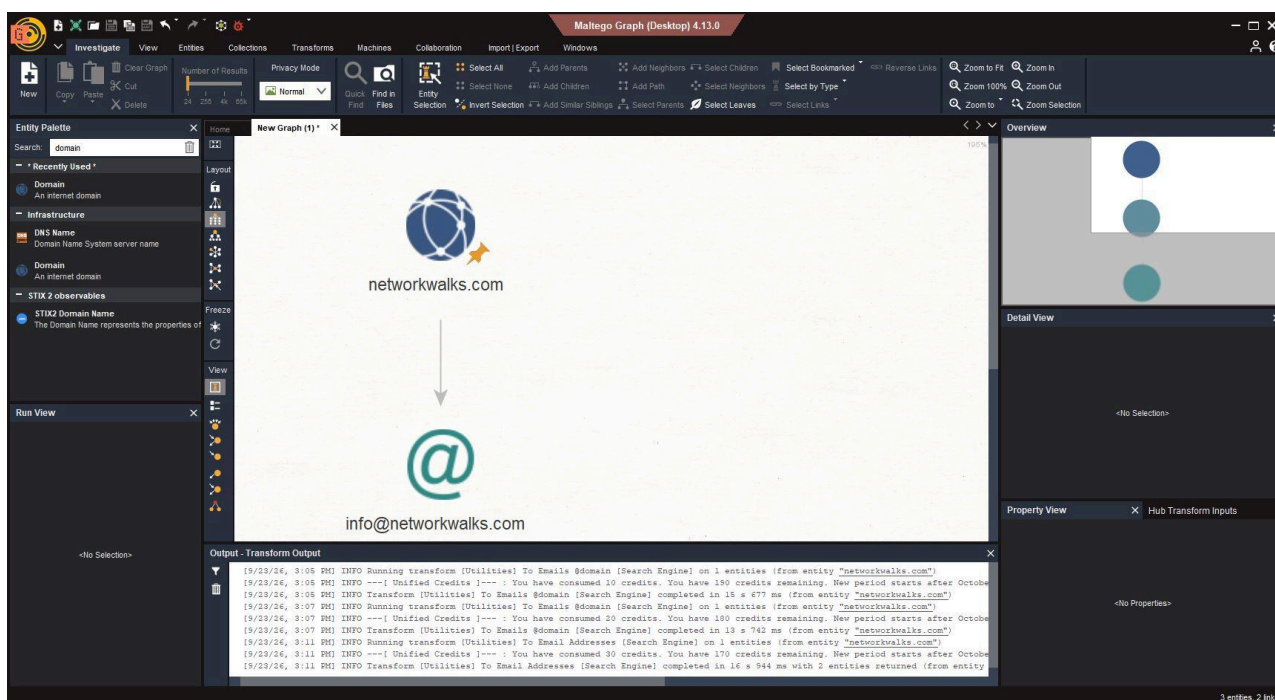
- b. Right-click the Domain entity, filter for email related transforms by searching “**email**,” select **[Utilities]** to **Email Addresses [Search Engine]** & run it.

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Penetration Testing Report



- c. The email address (**info@networkwalks.com**) related to **networkwalks.com** has now been harvested.



9. theHarvester

theHarvester is an open-source OSINT (Open Source Intelligence) footprinting tool designed to gather information about a target domain like emails, sub-domains, hosts, employee names, and open ports from different public sources like search engines and databases.

SOME KEY FINDINGS

- Emails,
- Sub-domains,
- Hosts,
- Employee names.

Penetration Testing Report

SOME COMMON ATTACK USES

- **Reveals related information:** An attacker would use this tool to find related open-source information about a target domain without touching it (to avoid raising any suspicions) in order to orchestrate well-tailored social engineering attacks.

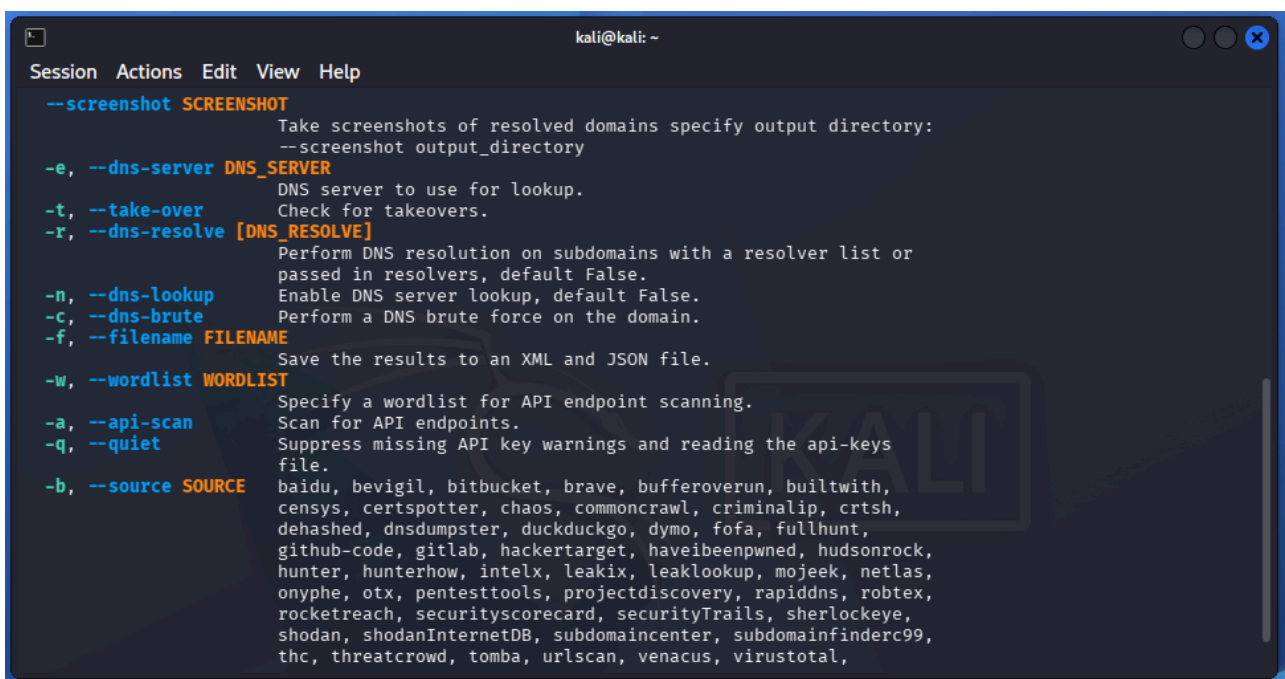
TASKS

This tool will be used to find email IDs & sub-domains related to Microsoft (microsoft.com) using theHarvester (in Kali Linux).

1. I found email IDs & sub-domains related to **microsoft.com** using the **theHarvester** tool in Kali with **Baidu**, with the **result limit set to 1000**.
 - a. In the Kali VM, open theHarvester tool from the menu. This launches in the Terminal interface.



- b. Read the usage instructions & guidelines.



- ```
theHarvester -d microsoft.com -l 1000 -b baidu
```

Screenshot showing the **theHarvester** command return in the Terminal CLI.

## CLI OUTPUT

[illegible]

## Penetration Testing Report

### CLI OUTPUT

```

[*] Target: microsoft.com

[*] Searching Baidu.

[*] No IPs found.

[*] No emails found.

[*] No people found.

[*] Hosts found: 4

c.urs.microsoft.com
hxd.research.microsoft.com
i.msdn.microsoft.com
social.microsoft.com
```

2. I found email IDs & sub-domains related to **microsoft.com** using the theHarvester tool in Kali with **all sources**, with the **result limit set to 50**.
  - a. In the terminal window, type the following command to start theHarvester & search for the required details using all sources:  

```
theHarvester -d microsoft.com -l 50 -b all
```

[illegible]

## CLI OUTPUT

[illegible]

## Penetration Testing Report

```
[*] Target: microsoft.com

Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bevigil search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bevigil.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

[!] Missing API key for Bitbucket.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bufferoverrun search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bufferoverrun.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to perform BuiltWith search for word: 'microsoft.com'
A Missing Key Error occurred in builtwith:
[!] Missing API key for BuiltWith.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process brave search for word: 'microsoft.com'
Error Message:
[!] Missing API key for Brave Search.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Censys API key is missing or invalid:
...
```

## Module 2: Scanning

Scanning is the phase that immediately follows footprinting and fingerprinting, focusing on discovering active hosts, open ports, and live services on a network. While fingerprinting identifies what a system is running, scanning maps out where the entry points are to identify reachable targets.

### Tools

- Zenmap

## Penetration Testing Report

### 1. Zenmap

Zenmap is the official GUI version of Nmap. It is an open-source multi-platform security scanner designed to provide an easier beginner-friendly graphical interface alternative to the CLI tool, Nmap, while adding advanced features.

#### SOME KEY FINDINGS

- Active devices,
- Open ports,
- Services and versions,
- Operating systems,
- Server hardware details.

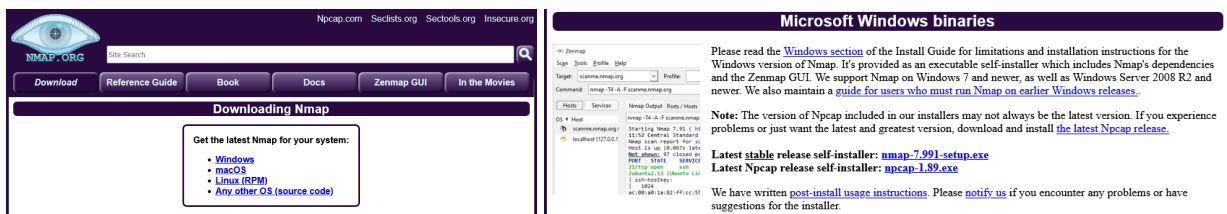
#### SOME COMMON ATTACK USES

- **Denial of Service (DoS) attacks:** Attackers may use this tool to find IP addresses and open ports, then use that data to orchestrate DoS attacks (like SYN flood attacks) in order to compromise availability.
- **Known Vulnerability Exploitation:** Attackers may collect device names, operating systems, and version numbers in order to exploit their known vulnerabilities to gain unauthorized access and steal data.

#### TASKS

This tool was used to find information on the available hosts inside my subnet, using the following steps.

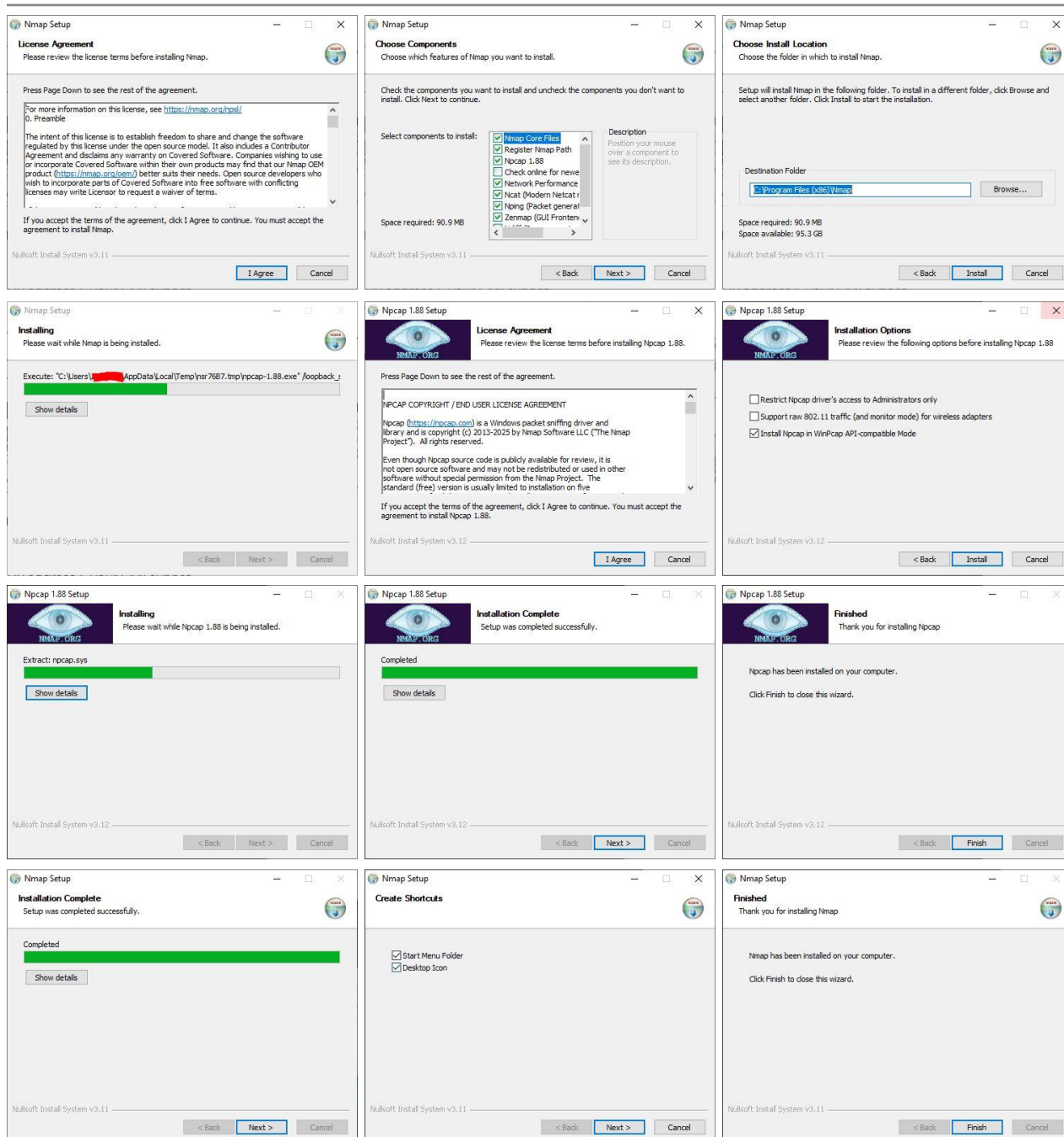
1. Download & install Zenmap from official website on your Windows PC
  - a. Visit and download the Zenmap installation file.



- b. Install the tool via the installation wizard.

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## Penetration Testing Report



## 2. Find your local IP address & your LAN subnet

- Open CMD & run the following command to find your PC's local IP address & your local LAN subnet.

`ipconfig`

## Penetration Testing Report

```

Administrator: Command Prompt
Microsoft Windows [Version 10.0.19044.7725]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

```

```

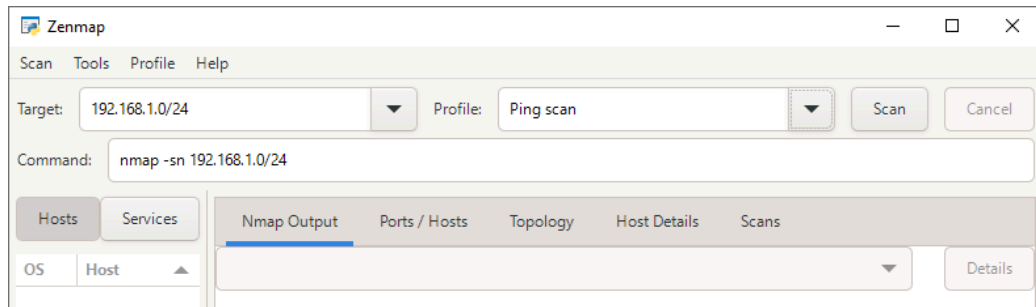
Connection-specific DNS Suffix . :
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::90e:5238:513c:2df8%23
IPv4 Address. : 192.168.1.13
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.1

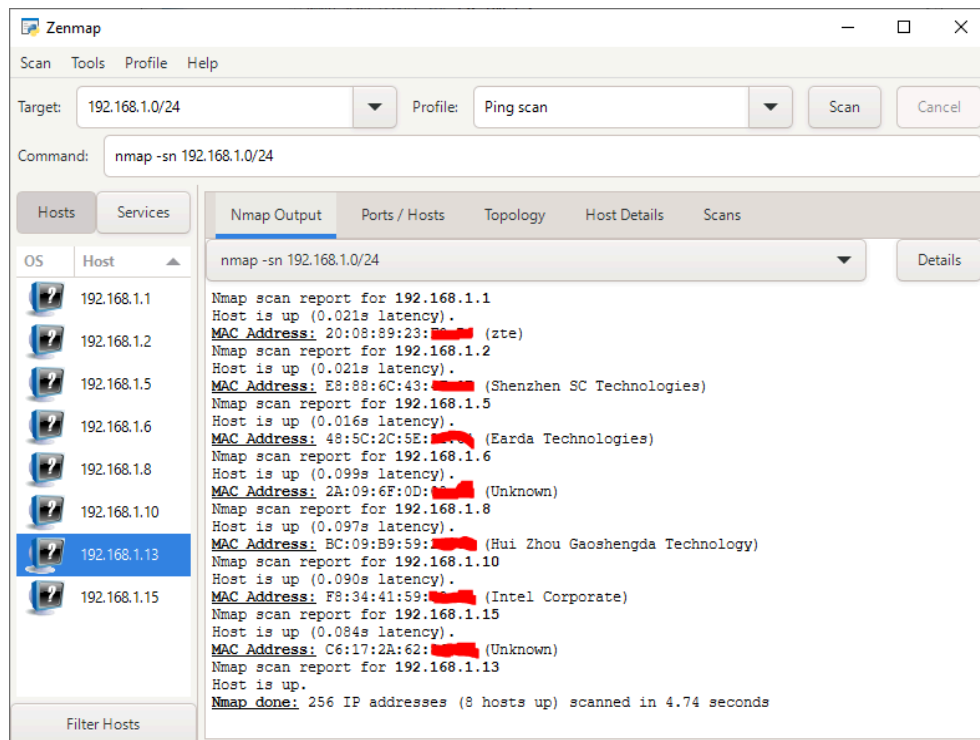
Ethernet adapter Ethernet 3:

```

3. Find the list of live hosts/PC's in your IP subnet
  - a. Open Zenmap, input the local LAN subnet, and select **Ping Scan** to find the list of live hosts in your subnet.



4. How many hosts are live on my subnet?
  - a. There are 8 live hosts on my subnet.



5. What are the IP addresses of the live hosts?
 

Including the host device:

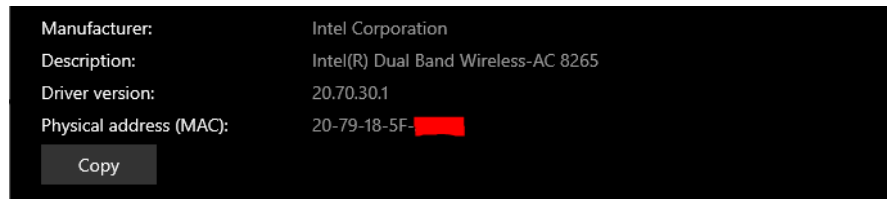
  - a. 192.168.1.1
  - b. 192.168.1.2

## Penetration Testing Report

- |                |                 |
|----------------|-----------------|
| c. 192.168.1.5 | f. 192.168.1.10 |
| d. 192.168.1.6 | g. 192.168.1.13 |
| e. 192.168.1.8 | h. 192.168.1.15 |

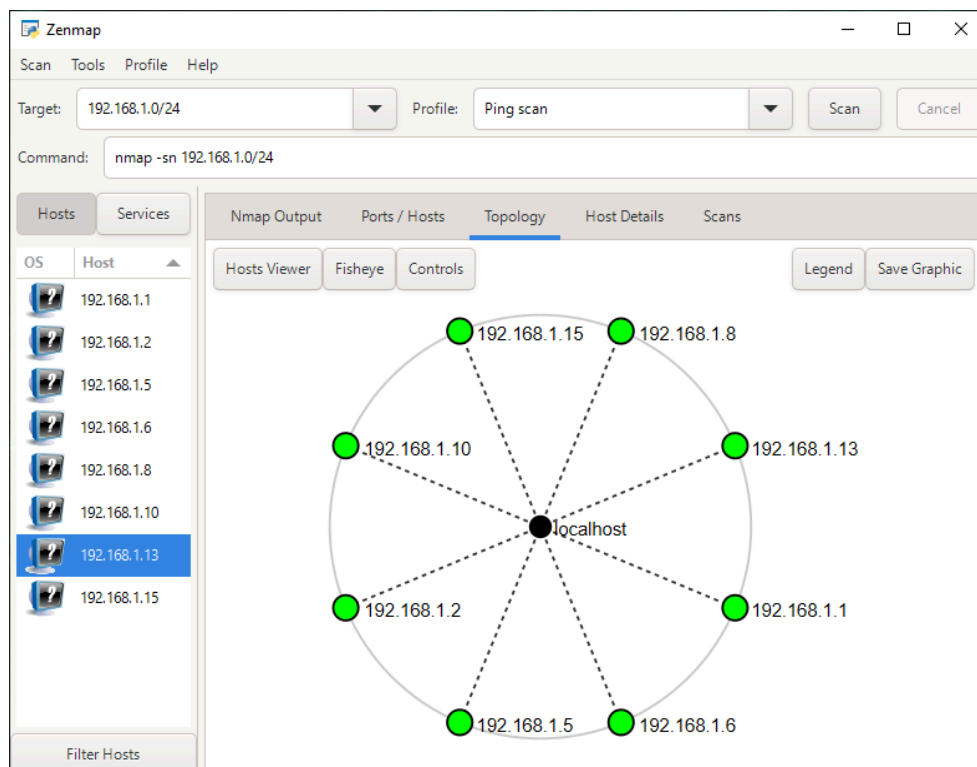
6. What are the MAC addresses of the live hosts?

I acquired my MAC Address from the network settings as it wasn't included in the Zenmap scan.



- |                      |                      |
|----------------------|----------------------|
| a. 20:08:89:23:XX:XX | e. BC:09:B9:59:XX:XX |
| b. E8:88:6C:43:XX:XX | f. F8:34:41:59:XX:XX |
| c. 48:5C:2C:5E:XX:XX | g. 20:79:18:5F:XX:XX |
| d. 2A:09:6F:0D:XX:XX | h. C6:17:2A:62:XX:XX |

7. Network Topology.



## Penetration Testing Report

